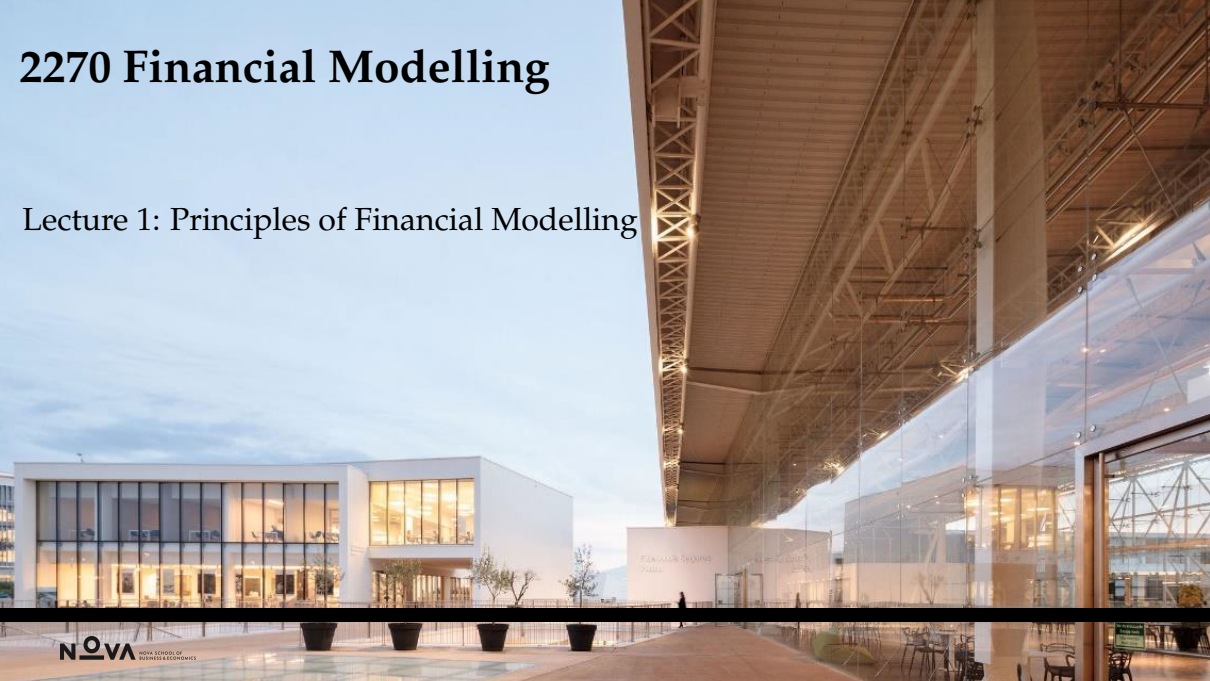
2270 Financial Modelling

Lecture 1: Principles of Financial Modelling



NOVA SCHOOL OF BUSINESS & ECONOMICS

What is Financial Modelling?

- Designing models for financial processes & decisions
  - Investment decisions
  - Asset valuation
  - Retirement savings

- Modelling in Python
  - Writing readable and efficient code
  - Manipulating, visualizing, and analyzing real-world data
  - Creating high-quality reports
  - Validating code

Course Schedule

- **Week 1:** Principles of financial modeling, introduction to Python
- **Week 2:** Bonds, data structures in Python and NumPy
- **Week 3:** Portfolio strategies, Pandas and APIs
- **Week 4:** Backtesting, portfolio optimization
- **Week 5:** Monte Carlo simulations
- **Week 6:** Mock exam + review

Recommended Textbooks

The lecture materials are self-contained, no textbook is required

- Benninga, Simon. Financial Modeling, 5th edition, MIT Press
- Hilpisch, Yves. Python for Finance: Mastering Data-Driven Finance, 2nd Edition. 2019
- Lewinson, Eryk. Python for Finance Cookbook, 2nd Edition. Packt Publishing, 2022.

Session Format

In class

- Discuss financial problems and associated models
- Run Python codes and analyze output
- Modify existing financial models for your tasks
- Weekly review quizzes

At home

- Improve your Python skills using DataCamp! Use this link to join the class. Make sure to use your novasbe.pt-email

I Have a Question?

- Teaching Assistants
  - Diogo António Rocha — diogo.rocha@novasbe.pt
  - Carlota Sousa Ferreira — carlota.s.ferreira@novasbe.pt
  - Maria Guimarães Neves — maria.neves@novasbe.pt
  - Tiago Frade Silva — tiago.silva@novasbe.pt
  - João Afonso Conde — joao.conde@novasbe.pt
- Moodle forum: post your questions about problem sets and slides
  ↪ Reply time within 48 hours
- Ask your questions during the breaks!

Changing sections

School Policy: Students are not allowed to change sections. Requests are to be made formally to the Masters team, not to the professor in charge.

- Rooms are at their capacity
- Mock exam will be assigned using sections

Grading

- Group Assignment 40% (Weeks 5–6)
  - Groups of 5-6 people: individual submissions are not allowed
- Final Exam 60%
  - 90-minute Wiseflow exam, multiple choice + open questions
  - December 15, 11:30 (make-up date is January 9, 16:00)
- Bonus point for completing all assignments on DataCamp

Can't I Just Use AI?

- AI has made *typing* code cheap — that is no longer the scarce skill
- Your job description changes: from **writer** of code to **editor** of code
- What the market now pays for:
  - **Specify**: translate a financial problem into a precise, implementable task
  - **Read**: understand code you did not write — line by line
  - **Verify**: check output against financial logic (does this Sharpe ratio pass the smell test?)
- None of this is possible without understanding the code and the finance underneath
  ↪ an editor who cannot read adds no value — and carries the liability

About Me

Something About You?

Principles of Financial Modelling

Modelling Overview

What is a Model?

- **Generic definition:** A representation of a thing, system, or theory such that characteristics of the original can be examined or changed
- **Business definition:** A computer program using structured logic to show the interactive relationships between dynamic variables, resulting in output to answer the questions of management

Modelling Overview

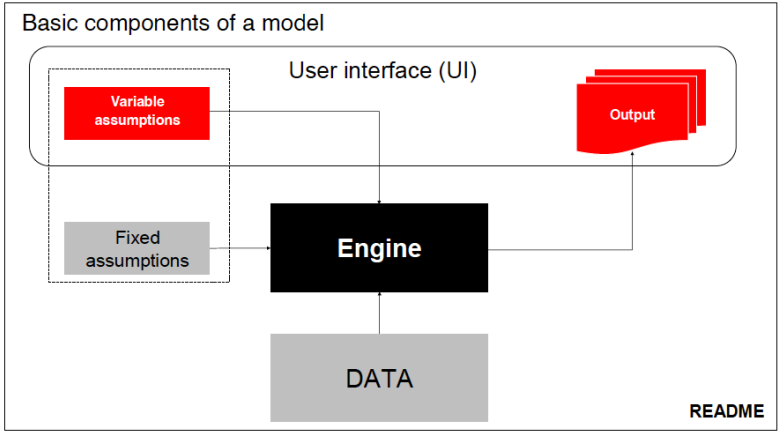
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Why Use a Model?

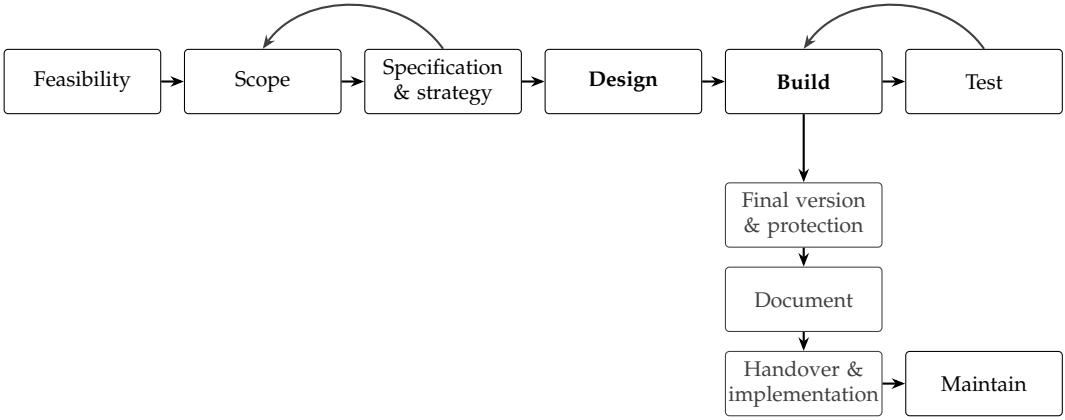
- **Accuracy/precision:** If an idea can be described, it can be modeled. Nothing is too conceptually complex or excessively detailed to be modeled
- **Efficiency:**
  - saves time on repetitive calculations
  - simplifies a business problem by breaking it into its component parts
  - documents the sources of data, assumptions, and methodologies in one place
- **Interactivity:** produces "what if" analyses to check the sensitivity of solutions to business changes – can be updated in real time

Modelling Overview

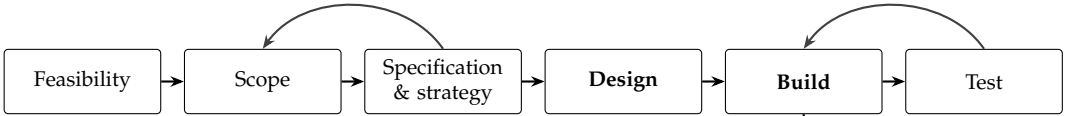


↪ Models may have **complex calculations**, but they always have **simple user interfaces**

The Financial Modelling Life Cycle

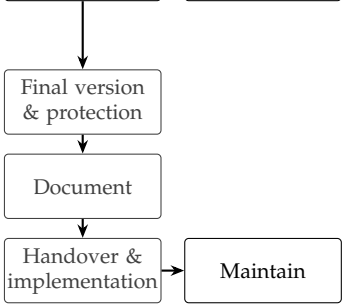


The Financial Modelling Life Cycle



- **Feasibility**: can and should the model be built? Evaluate available skills, people, budget, tools, timeframe, risks, stakeholder roles, and the cost of having (or not having) the model.
- **Scope**: state what is included *and* excluded. Define users, data responsibilities, intended use, maintenance and main cost drivers.

  Avoid: uncontrolled expansion of requirements that increases complexity, time, and cost.



How Do We Build a Model?

Structuring and Design

- Determine model's purpose
- Define output
- Identify required data
- Define flow and content of major sections
- Map the physical layout

⇓

! This step has high ROI: over 10 times return on time spent

- Model design can be a deliverable itself

Construction

- Establish assumptions
- Input required data
- Format model for easiest use based on established structure
- Test for errors

⇓

- With the proper structure and design, the process should be straightforward

Use

- Test logic and limits
- Update and modify the model as needed
- Create documentation

What is a Good Model?

It requires minimal time to understand

- clear logical flow
- can be broken into manageable components
- well-specified inputs
- complex functionality only where truly necessary

It is objectives-driven

- meets the decision-support needs of clients
- input variables and outputs are appropriate
- sufficient level of detail

It is free from errors

- no unintended consequences, no implicit assumptions
- correctly captures the relation between all variables
- easy to calibrate, i.e., make comparable with historical data

A good model should be coherent, well formatted, robust, simple and transparent, reusable and upgradeable

What is a Good Model? Process

Obtainable, accurate data

- obtainable: data is regularly published/produced (e.g., regulatory filings or investment reports) and thus updateable
- accurate: data source is not disputed

Objective vs. subjective

- assumptions and scenarios are reasonable and fact-based where possible
- rational basis for inclusion of data

Full documentation

- model purpose
- explanation of data flows and complex calculations
- data sources; how to update
- caveats and potential improvements

Financial Modelling in Finance

- **Investment banking:** Risk modeling, option pricing models
- **Insurance:** Insolvency models, actuarial models, risk models
- **Retail banking:** Funding and credit models
- **Corporate finance:** Capital budgeting models, cost of capital, financial statement analysis, governance models (SOX compliance testing)
- **Governments and institutions:** Econometric analysis-based models, macroeconomic models to analyze effects of government policy decisions on foreign exchange rates, interest rates, disposable income, and the gross national product

Introduction to Python

What's Python?

- Python is a general-purpose programming language conceived in 1989 by Guido van Rossum.
- Python is free and open source, with development coordinated through the Python Software Foundation.
- This is important because it
  - saves us money,
  - means that Python is controlled by the community of users rather than a for-profit corporation, and
  - encourages reproducibility and open science.

Python: Common Uses

- Python is a general-purpose language used in almost all application domains, including
  - AI and machine learning — Python is the language the entire AI stack is built in
  - scientific computing
  - web development
  - data engineering and automation
  - game development, multimedia, graphical interfaces
- It is used and supported extensively by tech firms including
  - Google, OpenAI, Anthropic, Netflix, Meta, Dropbox, Amazon

Python in the Financial Industry

- Python has ranked as the world's most popular programming language for years (IEEE Spectrum, Stack Overflow developer surveys)
- In finance: research, risk management, portfolio analytics, and trading pipelines run on Python — in many firms it has been replacing Excel for a decade
- And a new reason, unique to the AI era:
  - **Python is the language AI assistants write best** — it dominates their training data

  ↪ the better you read Python, the more leverage you get from AI tools

Installation

- Not Required!
- Google Colab: Jupyter notebooks

How to Learn Python

- Learning by doing — there is no other way
- DataCamp: your drill ground for syntax. Join here!
- Start by using AI to explain not solve
  - paste any line from the notebooks, ask “why does this work?” — infinitely patient tutor
  - but: code the AI writes *for* you is a skill you rent; code you understand is a skill you own